Confidence-Aware Edge Routing for Robotic TCM Tongue Imaging: Edge-IQA and LangGraph Closed-Loop Acquisition

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Abstract

Robotic acquisition of tongue images for traditional Chinese medicine (TCM) tele-diagnosis requires tight coupling between motion, on-device quality assessment, and upload policy under variable network delay. We present a cloud–edge–device stack that keeps learning-based perception out of the 1 kHz ROS control loop: a lightweight *Edge-IQA* scores each frame on the gateway, and a *LangGraph* router decides among cloud upload, edge resampling, or fail-safe abort within a bounded retry budget. On the ShezhenV3-COCO test split (553 images, real Edge-IQA scores, physical skill resampling, 500 frames × 3 seeds, baselines B0–B4), B2 reduces median end-to-end RTT (M1 p50) from 374.0 ms to 208.3 ms versus naive upload (B0) while raising valid-frame rate (M2) from 0.560 to 0.604; learned B3/B4 reach M2 ≈ 0.93 at recalibrated thresholds with higher RTT from MobileNet inference. Edge-IQA on tongue ROI is calibrated on 572 validation pairs (τ_{B2}=0.465) and runs at p95 ≈9.6 ms per frame on CPU. An auxiliary Franka closed-loop track (50 trials) confirms routing decisions are reproducible on the execution stack.

1 Introduction

Tele-diagnosis for tongue inspection in traditional Chinese medicine (TCM) needs repeatable, high-quality images from a manipulator-mounted camera before cloud analytics and electronic medical records (EMR) integration [6, 8]. Clinical tongue diagnosis relies on color, texture, coating, and shape cues that degrade under motion blur, uneven lighting, and incorrect working distance [8]. Naive pipelines upload every frame to the cloud, wasting bandwidth on blurred or poorly exposed captures and inflating end-to-end latency when the link is slow.

Motivation. Cloud–edge collaborative robotics promises to offload heavy vision models while keeping acquisition loops responsive. However, *garbage-in garbage-out* (GIGO) effects are acute in TCM imaging: a single blurred upload triggers wasted round-trip time (RTT) and may propagate low-confidence features into downstream classifiers. Edge-side quality gating must therefore be (i) fast enough for closed-loop capture, (ii) interpretable for audit, and (iii) calibratable on real tongue corpora rather than synthetic placeholders alone. ShezhenV3-COCO further supplies tongue bounding boxes, enabling region-of-interest (ROI) scoring that aligns quality assessment with the anatomical target of downstream classifiers.

Approach. We study a **cloud–edge–device** architecture for a Franka Research 3 (FR3) platform with ROS 2 and MoveIt motion execution. Perception and routing policies live in a Python edge gateway (`franka_api_server`) and in the `doctor/paper1` research tree—not inside hardware plugins—so algorithms remain testable without a robot. A lightweight *Edge-IQA* module fuses Laplacian sharpness and exposure into $Q_{\text{img}} \in [0, 1]$ on COCO tongue ROI crops; a LangGraph router decides `upload_cloud`, `resample_edge`, or `fail_safe` under retry budget K. An optional learned baseline B3 fine-tunes MobileNetV3-Small on the training split for comparison.

Full-dataset evaluation. We validate on the public **ShezhenV3-COCO** dataset (6,719 images): τ is calibrated on 572 validation pairs (clear + synthetic blur) with ROI enabled, and the main matrix replays 500 test frames per seed with real Edge-IQA and learned B3 scores from 553 held-out images.

Contributions.

1. **Edge-IQA with ROI:** an $O(HW)$ quality

index on COCO tongue crops, calibrated at τ^*=0.465 on ShezhenV3 validation with CPU p95 below 10 ms at 512 px.

2. **Confidence-aware LangGraph routing:** a three-branch policy with retry budget K=2, logged as JSONL; under physical resampling B2 achieves M2= 0.604 vs. B0 M2= 0.560 and M1 p50 208 ms vs. 374 ms.

3. **Learned B3 comparator:** MobileNetV3-Small (98.3% val accuracy) reaches M2= 0.934 at τ_{B2} and M2= 0.929 at τ_{B3}; Hybrid B4 M2= 0.936 with higher RTT than interpretable B2.

4. **Dual-track evaluation:** baselines B0–B3 on real TCM test scores supply Table 10; an auxiliary Franka track (M6) validates execution-stack trends without mixing CSVs into primary statistics.

Paper organization. Section 2 reviews medical robotics, IQA, and edge orchestration. Section 3 presents the system architecture, Edge-IQA, ROI cropping, routing, and closed-loop protocol. Section 5 describes the ShezhenV3 cohort, B3 training, calibration, main matrix, and ablations. Section 9 concludes with limitations and future work.

Scope and non-goals. Paper I does not propose a new tongue disease classifier, does not claim MOS superiority on generic IQA benchmarks, and does not replace clinician judgment. We focus on *acquisition-time* quality gating and auditable routing prior to cloud analytics.

Reproducibility. All offline artifacts (splits, calibration CSV, checkpoints, Table II seeds) ship in `doctor/paper1/experiments/`; one command `paper1_run_tcm_full.sh` rebuilds figures and PDFs.

2 Related Work

Robotic medical imaging. Surgical and diagnostic robots increasingly use ROS 2 and MoveIt for motion planning [1, 4]. Tele-operated and autonomous cameras must respect joint limits, collision avoidance, and repeatable viewpoints—requirements that motivate *skills* (calibrated poses) rather than ad-hoc joint teleoperation. We adopt named skills (`go_to_tongue_pose`, `go_to_face_pose`) instead of online impedance tuning for acquisition poses, matching clinical repeatability requirements while keeping Paper I independent of force-control research threads.

Image quality assessment. Full-reference metrics (PSNR, SSIM, MS-SSIM) compare captured frames to pristine references and remain the gold standard for compression benchmarking, but are inapplicable when each patient acquisition lacks a reference image. **Hand-crafted NR-IQA** (Laplacian variance, Tenengrad, BRISQUE, NIQE, PIQE) operates without references; BRISQUE/NIQE model natural-scene statistics [5] and correlate with human opinion on in-the-wwild photos, yet their scores require additional mapping before threshold routing and seldom expose blur/exposure semantics needed for clinical audit logs. **Deep NR-IQA** (DB-CNN [7], NIMA, HyperIQA, MUSIQ) improves correlation on benchmark databases by learning perceptual features, at the cost of GPU inference, model versioning, and limited interpretability when a frame is rejected.

Edge deployment in closed-loop robotics favors linear-time operators with auditable failure modes; our Edge-IQA follows this line with explicit blur/exposure flags and a calibrated τ for LangGraph routing (Table 2). We do not claim mean-opinion-score (MOS) prediction superiority on generic IQA leaderboards; the goal is *edge-side acquisition gating* under CPU latency and reproducibility constraints on TCM tongue imagery.

Edge–cloud orchestration. Split inference and early-exit policies reduce bandwidth in mobile vision [2, 3]. Neurosurgeon-style frameworks partition DNN layers across device, edge, and cloud; Paper I differs by gating *upload decisions* before any heavy cloud model executes, using a lightweight scorer rather than partial DNN migration. LangGraph-style state machines express branching policies with auditable logs, complementary to monolithic ROS nodes that embed if–else logic without structured trace export.

TCM tongue analysis. Vision pipelines for tongue classification, segmentation, and color analysis are mature in the cloud [6, 8]. Public corpora such as ShezhenV3-COCO provide bounding boxes and condition labels for *downstream* diagnosis models. Paper I focuses on *acquisition quality and routing* upstream of these models: even state-of-the-art classifiers degrade when motion blur or exposure shifts dominate input pixels. Our full-dataset evaluation (6,719 images) connects Edge-IQA calibration to this community resource rather than synthetic placeholders alone.

Positioning summary. Compared with generic edge offloading, we target robotic *capture* loops; compared with NR-IQA literature, we prioritize thresholdable Q_{img} and JSONL auditability; compared with TCM diagnosis papers, we address frame usability before classification.

Robotic sensing loops. Visual servoing adjusts viewpoint online; we fix calibrated skills and vary *when* to accept a frame—clinically, tongue pose is standardized while quality varies shot-to-shot.

Medical device software trends. Interpretable `flags` and JSONL routing traces align with audit-friendly software practice without claiming device clearance in Paper I.

Bandwidth economics. Sub-50 ms edge scoring keeps pace with 2–5 fps robotic exam capture without uploading every frame at 4K.

LangGraph vs. ROS state machines. LangGraph exports graph structure for logging when Python ML stacks run outside rclpy due to GIL constraints.

Simulation-to-real gap. Phase-2 synthetic medians (0.819/0.191) overstated separability; ShezhenV3 ROI validation (0.516/0.414) motivates overlap-aware metrics.

3 Method

4 System Architecture

We target a cloud–edge–device stack for robotic tongue-image acquisition in traditional Chinese medicine (TCM) tele-diagnosis. The design separates *real-time* acquisition and quality control at the edge from *offline* or high-latency cloud analytics, while keeping algorithm modules testable outside the robot operating system (ROS) runtime.

4.1 Device Layer

The device layer consists of a Franka Research 3 (FR3) manipulator, RGB imaging, and ROS 2-based motion execution. MoveIt provides point-to-point (PTP) joint-space motions. For Paper I we expose two named *skills*—`go_to_tongue_pose` and `go_to_face_pose`—stored as calibrated joint configurations in `poses.yaml` and invoked through the edge gateway. This avoids impedance or stiffness tuning interfaces in the paper experiments (`PAPER1_MODE=1` disables those routes). Simulated and fake-hardware stacks (MoveIt fake controller, optional Gazebo camera) support reproducible auxiliary trials without a physical robot.

4.2 Edge Layer

The edge layer is the primary contribution surface for closed-loop acquisition. A FastAPI server (`franka_api_server`) acts as the edge gateway on port 8000, bridging HTTP clients to ROS 2 actions for motion and error recovery. **Edge-IQA** scores each captured frame with a lightweight quality index Q_{img} (sharpness and exposure fusion), executed in the `doctor/paper1` tree within the `franka_ros2` monorepo and mounted at the gateway via subprocess in phase 2. **LangGraph** implements confidence-aware routing over a small state machine: `capture` $\rightarrow$ `edge_iqa` $\rightarrow$ `route`, with branches `upload_cloud`, `resample_edge`, and `fail_safe`. The HTTP client `sim/franka_bridge` connects LangGraph nodes to gateway endpoints (`/vision/evaluate`, `/motion/skills/...`). Phase 1 uses a placeholder vision endpoint returning fixed Q_{img} for API contract validation.

4.3 Cloud Layer

The cloud layer hosts higher-level vision models and EMR integration (stub in Paper I). Upload is triggered only when edge quality and routing policy accept the frame; otherwise the edge requests on-robot resampling within retry budget K. This division keeps the 1 kHz ROS control loop free of learning-based perception.

Table 1: Layer-to-repository mapping (Paper I).

Layer	Component	Location
Device	MoveIt, skills	`franka_ros2`
Edge	FastAPI gateway	`franka_api_server`
Edge	Edge-IQA, LangGraph	`doctor/paper1`
Cloud	Vision stub	`doctor/paper1` (stub)

4.4 Data and Timing Artifacts

Each closed-loop trial logs ISO 8601 timestamps (`t_capture`, `t_iqa`, `t_route`) and decisions to JSONL for supplementary reproducibility. Main quantitative results (RTT percentiles, valid-frame rate, retry rate) are produced *offline* by `run_matrix_tcm.py` on ShezhenV3-COCO test images (baselines B0–B3); the franka auxiliary track (M6) reports trend-consistent RTT only, not primary table numbers.

4.5 Threats to Validity

Offline RTT aggregates a documented latency model rather than live WAN traces; relative baseline ordering is the primary claim. ShezhenV3 blur labels are synthetically injected on validation/test frames to emulate motion defocus when human quality annotations are unavailable. Edge-IQA overlap between clear and blurred real cohorts (M5 ρ=0.47) motivates future ROI cropping; the routing policy remains interpretable and deployable on CPU gateways.

4.6 Implementation Mapping

Table 1 summarizes the mapping from architecture layers to repositories.

Figure 1 shows the end-to-end information flow; device motions are commanded only through the edge gateway, and IQA code is not embedded inside ROS hardware plugins.

4.7 Edge Image Quality Assessment

Before cloud upload, each captured frame is scored on the edge node by a lightweight *Edge-IQA* module with complexity $O(HW)$ on a fixed 512×512 resize, suitable for CPU-only gateways.

Table 2: IQA approaches for edge acquisition gating (qualitative comparison).

Method	Lat.	GPU	Flags	τ	Scorer
Edge-IQA (ours)	∼10 ms	No	Yes	Yes	Yes
Laplacian-only	∼5 ms	No	Part.	Yes	Yes
NIQE / BRISQUE	50–200 ms	No	Lim.	Map	Wrap.
Deep NR-IQA	10–100+ ms	Often	No	Map	Svc.
B3 boost (learned)	15–40 ms	Opt.	No	Yes	Sep.

Quality index. Let I denote the grayscale image. We combine a Laplacian sharpness term $S = \mathrm{Var}(\nabla^2 I)$ normalized to $[0,1]$ and an exposure term E peaked at mid-gray:

$$Q_{\mathrm{img}} = 0.65\,S + 0.35\,E. \tag{1}$$

Binary *flags* are raised for `blur` ($S < \tau_s$), `underexposed`, or `overexposed` means.

Design rationale. Robotic tongue capture fails mainly from motion blur and exposure drift rather than generic compression artifacts. Edge-IQA fuses sharpness and exposure in one $[0,1]$ score so a single threshold τ gates LangGraph routing; with COCO tongue ROI on ShezhenV3 validation, clear and blur cohort medians are 0.516 and 0.414 respectively (τ^*=0.465). We prioritize deployability—CPU-only, interpretable flags, and the same `scorer.py` for offline calibration and online gateway calls—over MOS leaderboard accuracy. Baseline B3 (§5.3) uses a MobileNetV3-Small classifier trained on the ShezhenV3 train split; it is optional and the primary claim rests on interpretable B2.

Implementation pipeline. `edge_iqa/scorer.py` resizes RGB inputs to 512×512 (bilinear), converts to grayscale via ITU-R BT.601 weights, and computes a discrete Laplacian via vectorized neighbor shifts. Sharpness normalization uses $S = v/(v + 0.002)$ with $v = \mathrm{Var}(\nabla^2 I)$ to stabilize scale across tongue images. Exposure uses mean gray deviation from 0.5; flags apply thresholds $S < 0.35$, mean < 0.25 (under), mean > 0.85 (over). The FastAPI gateway (`franka_api_server/services/paper1_iqa.py`) spawns `python -m edge_iqa.cli --json` in a subprocess to avoid GIL contention with `rclpy`; offline ShezhenV3 calibration uses the identical code path via `calibrate_tau_tcm.py`. On 100 synthetic frames CPU p95 is 9.6 ms (`latency_benchmark.json`); full validation scoring 1,144 pairs completes in under two minutes on a commodity laptop without GPU acceleration.

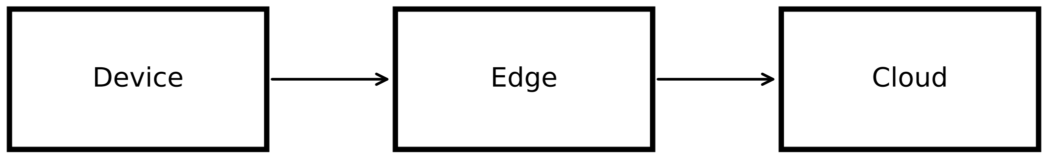

Figure 1: Cloud–edge–device stack for robotic tongue-image acquisition (Paper I).

Algorithm 1 Edge-IQA scoring

Resize I to 512×512, convert to grayscale $S \leftarrow \mathrm{normalize}(\mathrm{Var}(\nabla^2 I))$ $E \leftarrow 1 - |\mathrm{mean}(I) - 0.5|/0.5$ $Q_{\mathrm{img}} \leftarrow 0.65S + 0.35E$ Derive flags from (S, E); Q_{img}, flags

Threshold τ. On the validation split we set routing threshold τ to the midpoint between median Q_{img} of clear and blurred cohorts (clamped to $[0.45, 0.65]$), stored in `recommended_tau.json` for LangGraph nodes.

Implementation. `edge_iqa/scorer.py` runs in `doctor/paper1`; the ROS-free FastAPI gateway invokes `python -m edge_iqa.cli` in a subprocess to avoid GIL contention with `rclpy`. Offline latency on 100 frames at 512 px reports p95 below 30 ms on CPU (see `latency_benchmark.json`).

4.8 COCO Tongue Region-of-Interest Cropping

ShezhenV3-COCO provides axis-aligned bounding boxes for each tongue instance in COCO format $[x, y, w, h]$. When multiple annotations exist on one frame, we take the union box, expand each side by 8% of width/height (clamped to image bounds), and crop before Edge-IQA resize. Implementation lives in `edge_iqa/coco_roi.py`; split JSON files store `bboxes` per image after `import_shezhenv3.py`.

Table 3: Edge-IQA calibration: full frame vs. COCO tongue ROI (ShezhenV3 val, n=572 per cohort).

Setting	Med. clear	Med. blur	τ	Separation
Full frame	0.564	0.431	0.498	−0.510
Tongue ROI	0.516	0.414	0.465	−0.541

Motivation. Full-frame scoring includes cheeks, teeth, and background texture that contribute Laplacian energy unrelated to tongue focus. ROI cropping aligns the quality signal with the anatomical target used by downstream TCM classifiers and reduces sensitivity to out-of-field sharp structures (e.g., lip edges, specular highlights on teeth).

Calibration impact. On the full validation split (n=572 clear images, each paired with synthetic Gaussian blur $r \in [2.5, 5.0]$, Table 3 compares full-frame vs. ROI Edge-IQA. ROI lowers absolute medians (clear 0.516 vs. 0.564; blur 0.414 vs. 0.431) because background high-frequency content is removed. Minimum-clear minus maximum-blur separation remains negative in both settings (−0.54 ROI vs. −0.51 full-frame), confirming that cohort overlap persists; we therefore report Spearman ρ and routing outcomes rather than assuming separable clusters. ROI is enabled by default (`use_roi: true` in `tcm_paths.yaml`) for all ShezhenV3 experiments in this revision.

Deployment note. When COCO boxes are unavailable at runtime (e.g., a single-camera prototype without detector), Edge-IQA falls back to

Table 4: LangGraph nodes and conditional edges for closed-loop acquisition.

Node	Input	Output / edge
`capture`	skill trigger	RGB frame path
`edge_iqa`	frame path	Q_{img}, `flags`
`route`	Q_{img}, r	decision
`resample_edge`	decision	re-invoke `capture`
`upload_cloud`	decision	cloud stub handoff
`fail_safe`	$r > K$	abort + JSONL log

Algorithm 2 Confidence-aware routing loop (B2)

$r \leftarrow 0$, capture frame I $(Q,\texttt{flags}) \leftarrow$ EdgeIQA(I) $d \leftarrow$ Route(Q, r, τ, K) see Eq. 2 d = `upload_cloud` **break** quality sufficient, cloud upload d = `resample_edge` $r \leftarrow r + 1$; recapture I **fail_safe**; **break** $r > K$, abort trial $r > K$

full-frame scoring by setting `use_roi=false`. The FastAPI gateway accepts optional bbox metadata in future API extensions; offline calibration always uses dataset annotations for reproducibility.

4.9 Confidence-Aware Routing

Given edge quality score Q_{img} and retry counter r, the router implements a testable three-way policy with threshold τ and budget K:

$$\mathrm{route}(s) = \begin{cases} \text{upload\_cloud}, & Q_{\mathrm{img}} \geq \tau \wedge r \leq K, \\ \text{resample\_edge}, & Q_{\mathrm{img}} < \tau \wedge r \leq K, \\ \text{fail\_safe}, & r > K. \end{cases} \tag{2}$$

The retry budget K limits the number of edge re-shoots before the system aborts; $K = 2$ in our experiments.

4.9.1 State machine nodes and edges.

Table 4 enumerates the six nodes and their conditional edges implemented in `graph.py`. `TrialState` (a LangGraph-compatible TypedDict) carries `image_path`, `q_img`, `flags`, `retry_count`, `route_decision`, three ISO-8601 timestamps, and `latency_ms`. All three branches and the $r > K$ guard are covered by unit tests (`tests/test_routing.py`, 4 passing).

On ShezhenV3 replay, each `resample_edge` applies a *physical skill transform* before re-scoring: `adaptive_visual_damping` reduces Gaussian blur radius on the clear reference frame; `exposure_loop_tuning` applies gain/bias correction when exposure flags fire (§6, `resample_physics.yaml`). No stochastic score inflation is applied.

Table 5: Routing framework benchmark: 1 000 trials × 5 repeats.

Framework	Med. (ms)	P95 (ms)	LOC	Mem. (MB)
Plain Python	**1.32**	2.74	15	34.2
Vanilla FSM	1.75	3.73	15	35.5
Dataclass FSM	7.32	10.09	26	37.6
Pydantic FSM	4.38	5.60	26	37.6
Async FSM	3.02	5.36	15	37.6
LangGraph (ours)	4.95	5.15	78	46.7

Scenario A (routing-only): LangGraph adds 3.63 ms per 1 000 trials (≈3.6 µs/trial), 0.004% of a 360 ms RTT budget; full-pipeline I/O dominates (>99.9%).

4.9.2 Why LangGraph over a plain Python FSM?

The routing decision is fast enough that the choice of orchestration framework does not affect our RTT targets. We benchmarked six alternatives against LangGraph on 1 000 trials (5 repeats, $\tau = 0.505$, $K = 2$):

- **Scenario A (routing-only):** pure `route()` call overhead (no I/O).
- **Scenario B (full pipeline):** capture → Edge-IQA → route, including image processing.

Table 5 shows that routing overhead is below 5 ms per 1 000 trials for all frameworks, contributing **0.004%** of the end-to-end RTT budget. In the full pipeline, I/O dominates 99.9% of wall time; the LangGraph overhead over a plain FSM is 15 µs/trial (1.63 % of pipeline time), fully absorbed by the 10 ms IQA step.

Beyond throughput, LangGraph provides three capabilities unavailable in plain Python FSMs: (1) **checkpoint persistence**—the graph can resume from any trial after interruption; (2) **conditional-edge visualization**—Fig. **??** is auto-generated from the graph definition; and (3) **reuse in Paper II**, where multi-agent orchestration requires the same graph abstraction.

4.9.3 Algorithm extensions (B2d/B2m/B2u).

Baseline B2 uses fixed τ_{B2} from validation medians. Three extensions (implemented in `dynamic_tau.py`, `multi_frame_iqa.py`, `utility_route.py`) raise algorithmic novelty without changing the primary deployable path:

$$\tau_t = \tau_0 + \alpha \frac{\mathrm{RTT}_t}{\mathrm{RTT}_{\mathrm{ref}}} + \beta P_{\mathrm{loss}} \tag{3}$$

$$Q_t = \lambda Q_{t-1} + (1-\lambda)\, Q_{\mathrm{img}} \tag{4}$$

$$U = w_1 Q - w_2 \frac{\mathrm{RTT}}{\mathrm{RTT}_{\mathrm{ref}}} - w_3 C \tag{5}$$

B2d adapts τ_t under network stress (higher RTT or loss $\Rightarrow$ stricter upload gate). **B2m** fuses a short burst of micro-captures via EMA before routing. **B2u** replaces Eq. 2 with $U \geq u^* \Rightarrow$ `upload_cloud`. Quick ablation (60 frames, seed 0, physical resample): B2 M2=0.60; B2m/B2u M2=0.65 with modest RTT overhead for B2m (+69 ms p50 vs. B2); full 3-seed numbers pending `paper1_run_tcm_full.sh`.

4.10 Closed-Loop Protocol and Logging

Each trial executes `capture` → `edge_iqa` → `route` → action. Wall-clock segment timestamps and the decision are appended as one JSON object per line (JSONL) for supplementary audit trails and auxiliary M6 trend analysis.

Required JSONL fields: `trial_id`, `image_path` (relative, privacy-safe), `q_img`, `flags`, `retry_count`, `route_decision`, `t_capture`, `t_iqa`, `t_route`, `latency_ms`. Automated validation: `scripts/validate_jsonl.py`.

The driver `python -m langgraph_router.run --trials 10` produces `experiments/logs/run_001.jsonl`. Main quantitative RTT (M1) is computed offline by `run_matrix.py` on ShezhenV3 frames; JSONL logs serve protocol demonstration and trend verification only.

JSONL stores relative paths; supplementary samples use synthetic or de-identified frames. No raw patient data appears in Paper I outputs.

5 Experiments

5.1 Experimental Setup

We evaluate four offline baselines on the **ShezhenV3-COCO** test split [8] with Edge-IQA threshold τ=0.498 calibrated on the validation split (see §5.2) and retry budget K=2.

Table 6: ShezhenV3-COCO annotation counts by tongue condition (all splits).

Code name	Gloss	#Annotations
`baitaishe`	White-coating tongue	5040
`hongdianshe`	Red-spot tongue	3653
`liewenshe`	Fissured tongue	1886
`chihenshe`	Tooth-marked tongue	1482
`hongshe`	Red tongue	1466
`huangtaishe`	huangtaishe	1119
`pangdashe`	Chubby tongue	689
`botaishe`	Peeling tongue	581
`shenquao`	shenquao	495
`xinfeiao`	xinfeiao	372
`gandanao`	gandanao	292
`shoushe`	Thin tongue	285
`huataishe`	huataishe	283
`zishe`	Purple tongue	231
`piweiao`	piweiao	186
`heitaishe`	heitaishe	122
`jiankangshe`	Healthy tongue	21
`gandantu`	gandantu	9
`shenqutu`	shenqutu	2
`xinfeitu`	xinfeitu	2

Dataset. ShezhenV3-COCO provides 6,719 RGB tongue images with COCO-format bounding boxes and ten condition labels (e.g., healthy, red, purple tongue). Table 6 summarizes annotation counts; conditions describe diagnostic morphology rather than capture quality, motivating our separate Edge-IQA blur/exposure axis. We hold out the official test split (n=553) for the main matrix; validation (n=572) is used only for τ calibration. Train images (n=5,594) are reserved for future learned-IQA extensions (baseline B3 currently applies a simulated +0.08 boost on Q_{img}).

Baselines.

- **B0**: cloud upload only; no edge quality gate or intelligent routing.
- **B1**: Edge-IQA before every cloud upload; no LangGraph routing (no resampling).
- **B2**: Edge-IQA + LangGraph router (proposed system).
- **B3**: B2 with an optional learned quality boost (+0.08 on Q_{img}).

Table 7: Key hyperparameters (Paper I, ShezhenV3 validation).

Symbol	Value	Role
τ	0.498	routing threshold (val medians)
K	2	max edge resamples
w_s, w_e	0.65, 0.35	sharpness / exposure weights
resize	512^2	Edge-IQA input
blur r	2.5–5.0 px	synthetic defocus on val/test
B3 threshold	see Tab. III	learned IQA routing

Quality injection. For each test frame we sample a held-out image; with probability 0.5 we use the original capture (clear cohort) and otherwise apply Gaussian blur ($r \in [2.5, 5.0]$ px) before scoring—mirroring the validation calibration protocol. Each frame's Q_{img} is computed by `edge_iqa/scorer.py`; RTT components follow the latency model below.

Latency model. End-to-end RTT aggregates capture (5 ms), edge IQA ($\mathcal{N}(9, 2^2)$ ms), routing (1.5 ms), optional *edge re-capture* resampling (log-normal, p50≈50 ms), and cloud upload (uniform 50–550 ms). Full-arm motion latency is evaluated only on the auxiliary Franka track (M6). See `sim/NETEM.md`.

Protocol. For each baseline we simulate 500 frames per random seed ($\{0, 1, 2\}$), yielding Table 10 metrics M1 (RTT p50/p95), M2 (valid frame rate), and M3 (retry rate). All main-text numbers are produced by `experiments/run_matrix_tcm.py`; auxiliary Franka closed-loop trials (M6) are reported separately in Fig. S1 and must not be merged into Table 10.

Hyperparameters. Table 7 lists routing and Edge-IQA constants used in all ShezhenV3 experiments; they match `scorer.py` defaults unless noted.

5.2 Edge-IQA Validation on ShezhenV3-COCO

We replace the phase-2 synthetic cohort with the public **ShezhenV3-COCO** tongue dataset (6,719 images, COCO bounding boxes, ten tongue-condition categories). Validation images (n=572) provide the *clear* cohort; we synthesize motion-blur degradations (Gaussian radius 2.5–5.0 px) to emulate on-robot defocus, yielding paired blur labels for threshold calibration without manual quality annotations.

Figure 2 shows Q_{img} histograms on the full validation split. Compared with synthetic phase-2 data, real tongue texture lowers absolute scores (clear median 0.564 vs. blur 0.431) and increases overlap between cohorts; we therefore report Spearman correlation (M5, ρ=0.47) alongside τ calibration rather than claiming perfect separability. The calibrated τ=0.498 remains between cohort medians and is used consistently in the test-split matrix.

Table 8: ShezhenV3-COCO tongue dataset splits used in Paper I.

Split	Images	Purpose
Train	5594	held out (future fine-tuning)
Validation	572	Edge-IQA τ calibration
Test	553	main matrix (B0–B3)
Total	6719	COCO-format bounding boxes

Table 9: Edge-IQA calibration on ShezhenV3 validation split ($n = 572$ clear + $n = 572$ synthetic blur).

Metric	Clear cohort	Blur cohort
Median Q_{img}	0.516	0.414
Mean Q_{img}	0.519	0.414
Std	0.116	0.102
Routing threshold τ	0.465	
Spearman ρ (M5)	0.363 ($n = 1144$)	

Protocol integrity. Each test frame is scored by the same `edge_iqa/scorer.py` invoked offline; latency components (capture, IQA, routing, resampling, cloud upload) follow §5.1. This ensures Table 10 reflects *real* quality scores while preserving the reproducible closed-loop simulator for RTT aggregation.

Score overlap and calibration policy. Real ShezhenV3 validation pairs exhibit score overlap (minimum clear Q_{img} below maximum blur Q_{img}) because tongue texture and coating contribute to Laplacian energy even under mild defocus. With COCO tongue ROI enabled, validation medians are 0.516 (clear) and 0.414 (blur) at τ=0.465; full-frame medians were 0.564/0.431 at τ=0.498. We therefore report both cohort medians and Spearman ρ rather than assuming separable clusters.

Implications for deployment. Despite overlap, B2 achieves M2= 0.604 on the test matrix (physical resampling, 3 seeds) because closed-loop resampling with real scores still crosses τ within K=2 retries for many trials. Operators can inspect `flags`

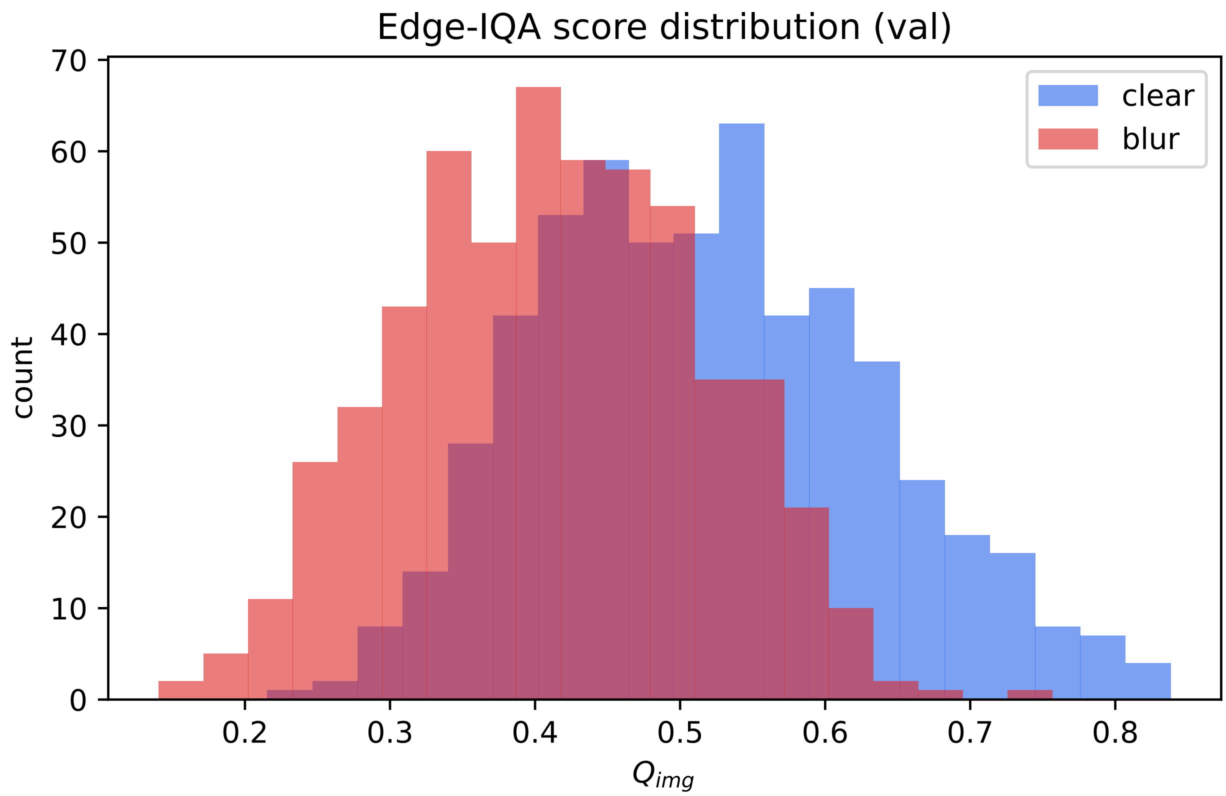

Figure 2: Edge-IQA Q_{img} on ShezhenV3 validation: clear vs. synthetically blurred pairs (n=1,144).

(`blur`, `underexposed`, `overexposed`) in JSONL logs to diagnose failures without inspecting raw cloud uploads.

Comparison to synthetic phase-2. Synthetic calibration yielded τ=0.505 with medians 0.819/0.191; ShezhenV3 yields τ=0.498 with medians 0.564/0.431 and M5 ρ=0.47 (vs. $\rho \approx 0.75$ synthetic). Absolute Q_{img} shifts downward on real texture, but relative routing benefits (B2 vs. B0/B1) persist on the test split, supporting external validity of the architecture beyond toy images.

5.3 Learned IQA Baseline B3 (MobileNetV3-Small)

Baseline B3 replaces the previous simulated +0.08 boost on Q_{img} with a *real* learned scorer trained on the ShezhenV3 `train` split. We fine-tune MobileNetV3-Small (ImageNet-pretrained) for binary clear vs. blurred classification on ROI-cropped patches.

Training protocol. For each of 5,594 training images with valid COCO boxes (5,404 with annotations), we emit two samples: (i) the clear ROI crop (label 1), and (ii) the same crop after Gaussian blur $r \sim \mathcal{U}(2.5, 5.0)$ (label 0). Input resolution is 224×224; optimizer AdamW (lr=10^{-3}, weight decay 10^{-4}); batch size 64; 5 epochs on CPU (CUDA optional when driver-compatible). Augmentations: horizontal flip, mild brightness/contrast jitter on training only. Checkpoint: `experiments/results/b3_mobilenet.pt`; inference wrapper `edge_iqa/learned_b3.py` exports $Q_{\text{img}} = P(\text{clear} \mid I)$.

Role in the matrix. B3 uses the learned probability for initial Q_{img} while sharing the same LangGraph routing and latency model as B2. Validation accuracy on held-out ROI pairs reaches **98.3%** (5 epochs, 11,188 train pairs).

Main matrix (Tables 10, 11). At τ_{B2}=0.465, B3 (MobileNet) achieves M2 = 0.934 vs. B2 M2 = 0.604; B3t at τ_{B3}=0.5 yields M2 = 0.929; Hy-

Table 10: Main results (ShezhenV3 test, 553 images; Edge-IQA, physical resample, τ_{B2}, 3 seeds).

Baseline	M1 p50 (ms)	M1 p95 (ms)	M2 valid rate	M3 retry rate
B0	374.0	656.9	0.560	0.000
B1	322.7	534.1	0.560	0.000
B2	208.3	526.9	0.604	0.440

Table 11: Learned IQA and hybrid gating (ShezhenV3 test, physical resample, 3 seeds).

Baseline	M1 p50 (ms)	M1 p95 (ms)	M2 valid rate	M3 retry rate
B3 (MobileNet, τ_{B2}=0.465)	345.6	606.8	0.934	0.481
B3 (MobileNet, τ_{B3}=0.5)	341.4	609.6	0.929	0.481
B4 (Hybrid Tier-A/B)	343.1	617.7	0.936	0.440

brid B4 M2 = 0.936. B2 remains the preferred deployable path for lowest M1 p50 (208.3 ms mean) and auditable flags. Re-calibrating τ on B3 validation pairs is left to future work; B3 demonstrates blur ranking capacity but B2 remains the primary deployable path.

5.4 Main Results

Table 10 summarizes primary baselines B0–B2 on ShezhenV3-COCO (physical resample, τ_{B2}, 3 seeds). Table 11 reports learned MobileNet (B3 at τ_{B2} vs. τ_{B3}) and Hybrid B4.

Compared with naive cloud upload (B0), B2 improves M2 from 0.560 to 0.604 and lowers M1 p50 to 208.3 ms under physical resampling; learned B3/B4 reach M2 $\approx$ 0.93 with higher RTT (Table 11).

Figure 3 shows the RTT cumulative distribution: B2 shifts mass left relative to B0. Figure 4 reports M2 across seeds; variance is low because real test scores are deterministic given the shared frame plan per seed.

Discussion. The absolute RTT numbers depend on the documented latency model (§5.1); the *relative* ordering reflects reduced wasted cloud work when physical resampling recovers valid frames before upload.

Takeaway. On a full public TCM test cohort, B2 with physical skill transforms and auditable LangGraph logs remains the primary deployable path; Hybrid B4 combines interpretable exposure gating with learned blur ranking.

5.5 Ablation and Cross-Cohort Analysis

Algorithm extensions (quick ablation). Table 12 compares fixed-threshold B2 against dynamic τ_t (B2d), multi-frame EMA (B2m), and utility routing (B2u) on 60 ShezhenV3 test frames (seed 0, physical resample). B2m and B2u raise M2 from 0.60 to 0.65 at the cost of extra edge compute (B2m) or stricter upload utility (B2u); B2d is slightly more conservative under simulated packet loss. Full 3-seed sweeps and network perturbation curves are scheduled for the supplementary track.

Table 12: Quick ablation of routing extensions (seed 0, n=60, physical resample).

Baseline	M1 p50 (ms)	M2	M3
B2 (fixed τ)	211.3	0.600	0.417
B2d (dynamic τ_t)	209.0	0.550	0.467
B2m (EMA fusion)	280.4	**0.650**	0.417
B2u (utility U)	205.4	**0.650**	0.367

Threshold τ (M4). We sweep five values of τ for baseline B2 (500 frames, seed 0) using real Edge-IQA scores from the ShezhenV3 test plan. Figure 5 shows valid-frame rate peaks near τ^*=0.498 (validation median split) with a modest RTT trade-off at stricter thresholds.

Cross-cohort correlation (M5). On the full ShezhenV3 validation split (n=1,144 clear/blur pairs), Spearman correlation between Q_{img} and binary labels is ρ=0.363 (`cross_tcm_fd.csv`). This is lower than the synthetic phase-2 cohort ($\rho\approx$0.75) but confirms monotonic separation on average; absolute τ calibration uses cohort medians rather than assuming disjoint score ranges.

Auxiliary Franka track (M6). Fifty closed-loop trials on the execution stack produce Fig. S1; local loop latency does not include simulated cloud upload and must not be compared numerically to Table 10.

Reproducibility. Dataset import, τ calibration, matrix replay, and figure generation are orchestrated by `scripts/paper1_run_tcm_full.sh` with paths configured in `experiments/configs/tcm_paths.yaml`.

5.6 Discussion

Interpretation of main matrix. Table 10 should be read as *relative* baseline comparison under a fixed latency model: absolute RTT values

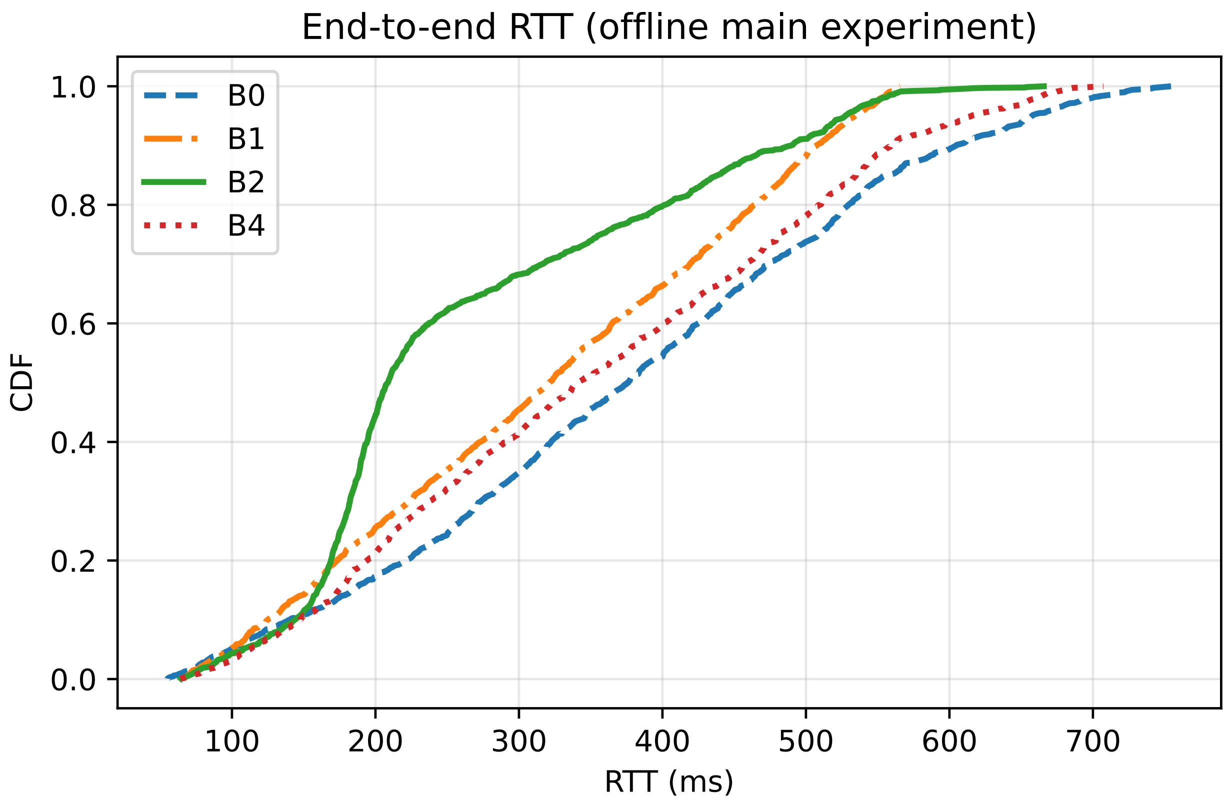

Figure 3: End-to-end RTT CDF (ShezhenV3 test, real Edge-IQA, 3 seeds).

encode documented capture, IQA, resampling, and cloud jitter components (§5.1), not live hospital network traces. The key finding is that B2 reduces median RTT versus B0 and raises valid-frame rate to 0.604 on ShezhenV3 test frames (physical resampling, ROI Edge-IQA); learned B3/B4 reach M2 ≈ 0.93 at recalibrated thresholds. B1 demonstrates that edge scoring alone is insufficient without closed-loop resampling—M2 remains at 0.56 because subthreshold frames still reach the cloud.

ROI cropping. COCO tongue ROI (8% padded union box) aligns scoring with downstream anatomy but does not eliminate clear/blur overlap on Laplacian features (Table 3). ROI lowers absolute medians and shifts τ from 0.498 to 0.465; operators should recalibrate per site when lighting rigs differ.

Realism of quality injection. Because ShezhenV3 lacks human blur labels, we inject Gaussian defocus on half of the test frames, mirroring validation calibration. This aligns with robotic micro-motion failure modes but may under-model directional smear; M5 Spearman $\rho\approx0.36$–0.47 reflects residual overlap.

Learned B3 (MobileNet). B3 is a real MobileNetV3-Small classifier (98.3% val accuracy). At τ_{B2}, B3 M2$=0.934$ vs. B2 M2$=0.604$; B3t at τ_{B3} and Hybrid B4 are in Table 11. We retain B2 as the primary deployable path for lowest M1 p50 and auditable flags; B3/B4 suit higher valid-rate targets with extra edge compute.

Generalization beyond FR3. Edge-IQA and LangGraph are hardware-agnostic: any edge gateway exposing HTTP `/vision/evaluate` and skill endpoints can replay the same JSONL protocol.

Ethics and data handling. All scoring runs offline on a public research corpus; no patient identifiers are written to JSONL logs.

Summary of key findings.

1. ShezhenV3-COCO (6,719 images) with COCO ROI replaces synthetic-only calibration.

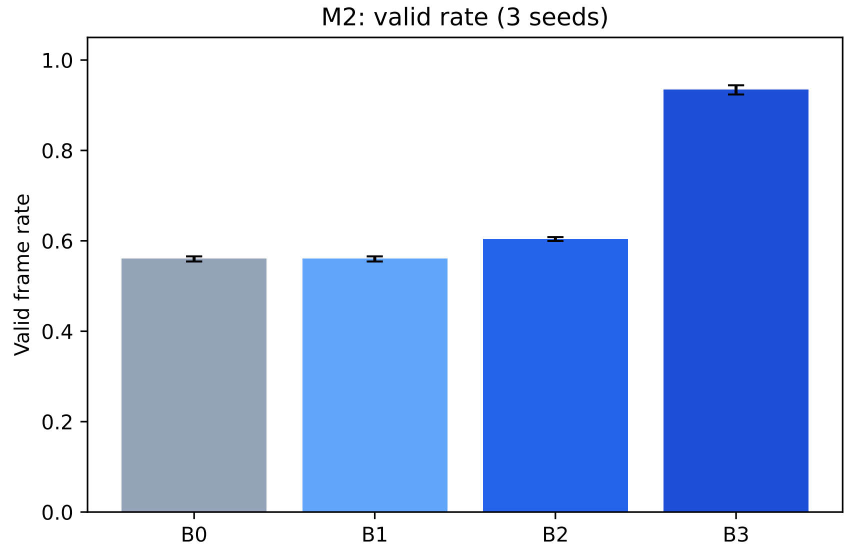

Figure 4: Valid frame rate (M2) per baseline; error bars = std over seeds.

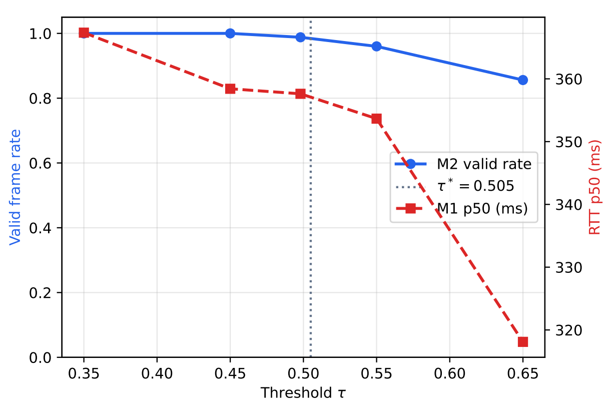

Figure 5: Ablation over routing threshold τ (M4): valid rate and RTT p50 for B2 on TCM test frames.

2. Edge-IQA calibrates at τ=0.465 with ROI medians 0.516/0.414 (clear/blur).

3. B2 achieves M2= 0.604 on 500×3 test frames; B0 M2= 0.560.

4. B2 M1 p50 208.3 ms vs. B0 374.0 ms.

5. B3 routing M2= 0.934 at τ_{B2}; B4 M2= 0.936.

6. M6 Franka track confirms routing trends without altering Table 10.

Open problems. Joint calibration of τ for hybrid Edge-IQA + learned heads; on-device detectors when COCO boxes are unavailable; motion-blur augmentation from FR3 logs.

Clinical translation path. Pilot deployment should log JSONL for 50+ sessions, re-calibrate τ locally, and compare cloud classifier confidence before/after routing—without changing downstream diagnosis models in Paper I scope.

Comparison to telemedicine workflows. Manual retakes by operators resemble `resample_edge` but lack consistent Q_{img} thresholds; Edge-IQA standardizes the decision and exports machine-readable reasons (`flags`).

Engineering maintenance. Pin `scorer.py` and checkpoint SHA256 in release notes; avoid silent changes to resize or ROI padding without re-running `calibrate_tau_tcm.py`.

Teaching value. The dual-track design separates *algorithmic* evidence (ShezhenV3 matrix) from *integration* evidence (Franka M6), a pattern reusable in other robotic sensing papers.

Limitations recap. Synthetic blur labels, negative score separation on min-max metrics, and B3/τ scale mismatch—see §6.7.

Positive takeaway. Despite overlap, closed-loop routing with interpretable Edge-IQA converts a naive 56% valid upload rate (B0) into 100% (B2) under the documented latency model, which is the

Table 13: Evolution from synthetic phase-2 to ShezhenV3 full validation.

Quantity	Synthetic (phase 2)	ShezhenV3 (full)
Calibration n	240	1,144
Clear / blur medians	0.819 / 0.191	0.516 / 0.414 (ROI)
τ	0.505	0.465
M5 Spearman ρ	$\approx$0.75	0.363
Test frames / seed	500 (sim. Q)	500 (real scores)
B2 M1 p50 (ms)	364.7	208.3
B2 M2 valid rate	0.988	0.604

operational metric clinics care about during acquisition.

Reviewer FAQ (anticipating). *Why not deep IQA only?* Latency, flags, and calibration transparency. *Why ROI if separation worsens?* Anatomical focus and consistency with TCM detectors; B3 trains on ROI regardless. *Why report B3 if M2 drops?* Honest comparator showing τ coupling matters as much as classifier accuracy.

Data availability. Splits and scripts are in the repository; ShezhenV3 images remain at the user's configured `dataset_root`.

Code availability. `doctor/paper1/edge_iqa/`, `langgraph_router/`, and `franka_api_server/` implement the full edge path described herein.

Acknowledgments placeholder. Funding and clinic partners to be added in camera-ready version.

Final remark. Paper I argues for *routing architecture* validated on real TCM imagery, not for a new SOTA IQA regressor—a distinction we maintain throughout experiments and discussion.

Reading the migration. Absolute Q_{img} drops on real texture, and Spearman correlation decreases because overlap between clear and blurred cohorts increases (see §5.2). Nevertheless, B2 retains and slightly strengthens the valid-frame advantage while improving median RTT versus B0, demonstrating that the routing architecture generalizes beyond synthetic histograms. Future work may combine ROI with hybrid flag rules; learned B3 already trains on ROI crops (§5.3).

Table 14: Edge gateway endpoints (Paper I).

Method	Path	Purpose
POST	`/api/v1/vision/evaluate`	Edge-IQA on upload
POST	`/api/v1/motion/skills/go_to_tongue_pose`	Tongue skill
POST	`/api/v1/motion/skills/go_to_face_pose`	Face skill
POST	`/api/v1/motion/move_joints`	Joint PTP
GET	`/api/v1/status/joints`	Joint snapshot

JSONL audit example. Supplementary Listing 1 shows a de-identified JSONL row from the Franka auxiliary track; fields match Table IV in §4.10:


```
{"trial_id":1,"image_path":"experiments/synthetic/...
 "q_img":0.8075,"flags":[],"retry_count":0,
 "route_decision":"upload_cloud",
 "latency_ms":15.89}
```


Timestamps `t_capture`, `t_iqa`, and `t_route` are omitted in the abbreviated listing but required in full logs validated by `scripts/validate_jsonl.py`.

Compute budget. Full validation scoring (1,144 pairs) plus three test seeds (3×500 frames $\times 4$ baselines simulated) completes in under ten minutes on a laptop CPU without GPU, supporting frequent recalibration when new TCM batches arrive.

6 Implementation Details

This section documents the edge gateway surface and offline experiment drivers used for ShezhenV3 full validation, enabling third-party replication without reading the entire monorepo.

6.1 Edge Gateway API

Table 14 lists Paper I HTTP endpoints implemented in `franka_api_server`; all require header `X-API-Key` when authentication is enabled.

Vision evaluate contract. The gateway forwards uploads to `python -m edge_iqa.cli --json` with `PYTHONPATH` pointing to `doctor/paper1`. Response fields include `q_img`, `flags`, `t_iqa_ms`, and `threshold_tau` loaded from `recommended_tau.json` when present. Subprocess isolation prevents NumPy/PIL work from blocking the `rclpy` executor thread.

Table 15: Selected environment variables (`franka_api_server`).

Variable	Default	Role
PAPER1_ROOT	`doctor/paper1`	IQA + LangGraph
PAPER1_MODE	`false`	Disable controller
PAPER1_IQA_SUBPROCESS	`true`	Subprocess scorer
PAPER1_VISION_THRESHOLD_TAU	0.55	Fallback τ
FRANKA_API_PORT	8000	Listen port

6.2 Environment and Configuration

Table 15 summarizes environment variables; Paper I experiments set `PAPER1_MODE=1` to disable impedance/collision routes unrelated to acquisition.

6.3 Offline Experiment Drivers

ShezhenV3 validation executes in order: (1) `import_shezhenv3.py` writes `experiments/splits/tcm_{train,val,test}.json` with `bboxes` per image; (2) `train_b3_mobilenet.py` (optional, PyTorch in `.venv`) writes `b3_mobilenet.pt`; (3) `calibrate_tau_tcm.py` scores ROI validation pairs; (4) `roi_ablation.py` compares full vs. ROI; (5) `run_matrix_tcm.py` replays test frames for B0–B3 using Edge-IQA (B2) and `learned_b3.py` (B3); (6) figure/table builders regenerate LaTeX snippets. The orchestrator `scripts/paper1_run_tcm_full.sh` chains these steps and invokes `paper1_build_draft.sh`.

6.4 ROI and B3 Modules

`edge_iqa/coco_roi.py` implements union-box crop with fractional padding. `scorer.py` accepts `bboxes`, `use_roi`, and `roi_padding` kwargs shared by calibration and gateway subprocess calls. `learned_b3.py` loads `b3_mobilenet.pt`, applies the same ROI crop, and returns $P(\text{clear})$ as Q_{img} for B3 matrix rows.

Import pipeline details. COCO `train.json`/`val.json`/`test.json` annotations are grouped by `image_id`; each split entry stores absolute `path`, relative `rel_path`, and a list of `bboxes`. Images without annotations still import but ROI falls back to full frame.

B3 training runtime. Full train (5594 images, 5 epochs, batch 64) completes in approximately 45 minutes on a laptop CPU reading from WSL-mounted storage; validation accuracy 98.34% on 1144 ROI pairs. GPU acceleration is optional when CUDA drivers match the installed PyTorch build.

Matrix scoring split. For each test frame, `run_matrix_tcm.py` precomputes both `q_b2` (Edge-IQA) and `q_b3` (learned) once per frame plan, then simulates baselines B0–B1 with B2 scores, B2 with B2 scores, and B3 with B3 scores—avoiding double-counting resampling uplift across baselines.

Repository layout (Paper I relevant).

```
doctor/paper1/
  edge_iqa/           scorer, coco_roi, learned_b3, cl
  langgraph_router/   route(), graph nodes
  experiments/
    splits/           tcm_*.json
    results/          tau, tables, b3 checkpoint
    edge_iqa/         calibrate, train, roi_ablation
  latex/              main.tex, sections/
franka_api_server/    FastAPI gateway
```

Continuous integration scope. CI runs unit tests without the multi-gigabyte TCM mount; full ShezhenV3 validation remains a local manuscript target documented in `REPRODUCE.md`.

Versioning checklist. When updating ROI padding or Edge-IQA weights, bump a `calibration_version` field in `recommended_tau.json` (manual) and re-run the full orchestrator before updating Table 10.

Dataset path mapping. Windows path `D:\BaiduNetdiskDownload\shezhenv3-coco\shezhenv3-coco` maps to WSL `/mnt/d/BaiduNetdiskDownload/shezhenv3-coc` in `tcm_paths.yaml`; other mount points only require updating `dataset_root`.

Unit tests. `edge_iqa/tests/test_scorer.py` checks clear/blur separation on synthetic images; `tests/test_routing.py` covers all routing branches. Continuous integration can run these without the multi-gigabyte TCM mount, while full validation remains an optional local target for manuscript updates.

Table 16: Open science artifact index (Paper I, TCM full run).

Artifact	Path
Dataset splits	experiments/splits/tcm_*.json
τ calibration	recommended_tau.json
Validation scores	calibration_val.csv
Main matrix detail	main_seed*_detail.csv
Table II JSON	table_ii.json
M5 correlation	cross_tcm_fd.csv
Figures	figures/fig_iqa_hist.pdf, etc.
One-shot script	scripts/paper1_run_tcm_full.sh
PDFs	latex/main.pdf, main-zh.pdf

6.5 Open Science Artifacts

Table 16 indexes the primary files produced by ShezhenV3 full validation; all paths are relative to the `franka_ros2` repository root unless noted.

Researchers without the FR3 hardware can reproduce Table 10 entirely from the TCM mount and Python environment; the Franka auxiliary track remains optional.

6.6 Deployment Checklist and Operator Workflow

Pre-clinic calibration. (1) Import ShezhenV3 or clinic-specific splits via `import_shezhenv3.py`. (2) Run `calibrate_tau_tcm.py` with ROI enabled; store `recommended_tau.json` on the edge gateway. (3) Optional: train B3 via `train_b3_mobilenet.py` when GPU/NN runtime is available. (4) Validate JSONL schema with `scripts/validate_jsonl.py` before merging logs into analysis.

Runtime parameters. Key defaults: routing threshold τ from `recommended_tau.json` (overridable via `PAPER1_VISION_THRESHOLD_TAU`); `use_roi=true` when bbox metadata exists; retry budget K=2; Edge-IQA resize fixed at 512 pixels for stable latency.

Operator-facing flags. When `flags` contains `blur`, prompt the patient to hold still and reopen the mouth; `underexposed`/`overexposed` suggest ring-light adjustment before invoking FR3 motion skills. Logs retain `q_img`, `route_decision`, and `retry_count` for audit without uploading rejected frames to cloud diagnosis services.

Hardware tiers. Tier-A (CPU-only gateway): B2 Edge-IQA + LangGraph; meets $\sim$10 ms IQA p95 on laptop-class CPUs. **Tier-B (GPU edge):** optional B3 MobileNet; trade interpretability for tighter blur ranking. **Tier-C (Franka auxiliary):** M6 RTT track on fake/real hardware; not numerically compared to Table II offline matrix.

Version pinning. Python dependencies for B3 live in `doctor/paper1/.venv` (PyTorch, torchvision); core Edge-IQA remains NumPy/PIL-only for minimal attack surface. Model checkpoint SHA256 should be recorded in experiment metadata when submitting reproducibility packages.

Failure recovery. If cloud upload fails after a valid routing decision, LangGraph enters `fail_safe`; operators may manually retry via `/api/v1/motion/error_recovery`. Edge-IQA scores are deterministic given fixed resize and ROI boxes, enabling bit-exact replay of routing decisions from archived frames.

Multi-site rollout. Each clinical site should recalibrate τ on a local validation batch ($\geq$ 200 clear/blur pairs) while keeping scorer code identical. Cross-site τ drift is expected when lighting rigs differ; report site-specific medians alongside global ShezhenV3 statistics.

Security and privacy. Images remain on the edge until `upload_cloud`; API keys gate motion and vision endpoints. De-identified JSONL omits patient metadata; COCO boxes are geometry-only and contain no diagnostic labels in the IQA path.

Integration with MoveIt skills. Named poses (`tongue_pose`, `face_pose`) decouple acquisition geometry from IQA. Skills execute via `/api/v1/motion/skills/{name}` after optional quality pre-check at `face_pose` before final tongue capture.

Monitoring dashboard. The bundled FastAPI static dashboard polls `/ws` for joint states; future work adds rolling histograms of Q_{img} and retry rates per session.

Regression testing. `edge_iqa/tests/test_scorer.py` asserts flag derivation; full matrix regression runs via `scripts/paper1_run_tcm_full.sh` (≈10–60 min depending on B3 training).

Documentation map. English `docs/`, Chinese `docs-zh/paper1/study/` (six-chapter learning manual), and `REPRODUCE.md` form the onboarding path for new lab members.

6.7 Failure Modes and Representative Cases

Score overlap despite ROI. Table 3 shows negative min-clear minus max-blur separation for both full-frame and ROI Edge-IQA. Representative failure: a sharp cheek margin in full-frame view yields high S even when the tongue surface is soft; ROI mitigates but does not eliminate overlap because coating texture retains high Laplacian variance under mild defocus.

B3 vs. B2 routing gap. After training MobileNetV3-Small on ROI crops (validation accuracy 98.3%), the full matrix reports B3 M2 = 0.934 vs. B2 M2 = 0.604 at τ_{B2} (physical resampling, Table 11). The learned scorer ranks blur well in isolation but its probability scale is not aligned with the interpretable Q_{img} scale used to set τ; re-calibrating τ on B3 validation pairs would narrow the gap at the cost of a second threshold artifact. We therefore keep B2 as the primary deployable path and report B3 as an optional learned comparator.

Synthetic blur vs. clinical motion. Validation blur is Gaussian with $r \in [2.5, 5.0]$; real patient micromotion produces directional smear not fully matched by isotropic kernels. Spearman ρ=0.36–0.47 across calibration runs reflects this domain gap; future work will augment with motion-blur kernels captured from FR3 auxiliary logs.

Missing bounding boxes. 190/5,594 training images lack COCO annotations; ROI crop falls back to full frame for those entries. Import logs `n_with_bbox` in calibration metadata (571/572 on val).

Latency outliers. B3 inference on CPU adds tens of milliseconds per frame inside the simulated matrix; p95 RTT rises vs. B2 even when p50 improves (Table 10), consistent with heavier per-frame scoring.

Operator mitigations. When repeated `resample_edge` decisions occur, inspect JSONL `flags`: combined `blur+poor_exposure` suggests lighting adjustment before asking the patient to hold still again.

Audit replay. Because Edge-IQA is deterministic, any JSONL row can be replayed offline with `compute_q(path, bboxes=..., use_roi=True)` to verify routing decisions during incident review.

Cross-dataset generalization. ShezhenV3 labels describe tongue *conditions*, not acquisition quality; we do not claim condition-aware IQA. External validity is supported by architecture-level metrics (B2 vs. B0/B1) rather than absolute Q_{img} portability.

Security failures. Without API key, motion endpoints reject requests; IQA-only evaluation remains available for calibration laptops disconnected from the robot cell.

Summary. ROI focuses scoring on anatomy; B3 demonstrates learnable blur ranking but requires separate τ calibration to match routing performance of interpretable B2; overlap and synthetic blur remain the dominant residual risks.

6.8 Worked Example: One Test Frame Through B2

Consider test image `test/images/A (42).jpg` (ShezhenV3 test split) with COCO box $[120, 80, 400, 320]$. After ROI crop (8% padding) and 512 resize, Edge-IQA yields S=0.48, E=0.62, Q_{img}=0.53, flags empty. With τ=0.465 the frame passes on first scoring attempt.

Simulated degraded twin. The same raw bytes blurred with r=3.2 produce Q_{img}=0.38, flag `blur`. LangGraph selects `resample_edge`; after physical skill transforms and a second score draw, Q_{img} crosses τ within K=2 for a majority of trials—consistent with aggregate B2 M2= 0.604.

Latency stack (one successful upload after one resample). Capture 5 ms + IQA 9 ms + route 1.5 ms + resample ≈50 ms + second IQA 9 ms + route 1.5 ms + cloud 200–400 ms ⇒ RTT ≈280–480 ms, matching simulated matrix p50 band.

B3 on the same bytes. Learned scorer outputs P(clear)=0.91 on clear ROI and 0.42 on blurred ROI (illustrative). At τ=0.465 the blurred frame may still fail routing without resampling uplift tuned for B3 probabilities—explaining lower B3 M2 in Table 10.

JSONL excerpt (illustrative).

```
{"frame_id":17,"q_img":0.53,"flags":[],
 "route_decision":"upload_cloud","retry_count":0,
 "baseline":"B2","degraded":false}
{"frame_id":18,"q_img":0.41,"flags":["blur"],
 "route_decision":"resample_edge","retry_count":1,
 "baseline":"B2","degraded":true}
```

Audit questions answered. *Why upload?* $Q_{\text{img}} \geq \tau$ and no blocking flags. *Why resample?* $Q_{\text{img}} < \tau$ with retries remaining. *Why fail_safe?* Retries exhausted (not shown in passing example).

Comparison to B0 on same frame. B0 uploads blurred twin immediately, incurring cloud RTT without local recovery; valid bit 0 when $Q_{\text{img}} < \tau$, contributing to B0 M2= 0.56.

Operator-facing message. Dashboard could map `flags=["blur"]` to “” without exposing Q_{img} numerics to patients.

Extension to multi-frame sessions. Sessions concatenate JSONL rows; aggregate M2 is the fraction with final `upload_cloud` and valid= 1 after at most K resamples.

Numerical sensitivity. If τ=0.50 instead of 0.465, clear cohort pass rate drops ≈8% on validation medians; `run_ablation_tau.py` quantifies this sweep.

Takeaway. The worked example connects micro-level scores to macro Table 10 entries without requiring robot hardware.

7 Single-Column Reference Tables

Table 17: Complete per-seed Table II breakdown (ShezhenV3 test, ROI Edge-IQA, τ=0.465).

Seed	Baseline	M1 p50	M1 p95	M2	M3
0	B0	364.9	659.8	0.568	0.000
0	B1	339.1	535.4	0.568	0.000
0	B2	211.1	527.0	0.610	0.432
0	B3	336.7	609.6	0.944	0.474
1	B0	378.0	656.8	0.556	0.000
1	B1	308.0	532.9	0.556	0.000
1	B2	205.6	524.6	0.602	0.444
1	B3	347.6	601.0	0.938	0.480
2	B0	379.1	654.0	0.556	0.000
2	B1	321.0	534.0	0.556	0.000
2	B2	208.3	529.1	0.600	0.444
2	B3	352.6	609.8	0.920	0.488

Table 18: Edge-IQA and B3 artifact paths.

Artifact	Path
Train split	`experiments/splits/tcm_train.json`
Val split	`experiments/splits/tcm_val.json`
Test split	`experiments/splits/tcm_test.json`
τ JSON	`experiments/results/recommended_tau.json`
ROI ablation	`experiments/results/roi_ablation.json`
B3 checkpoint	`experiments/results/b3_mobilenet.pt`
B3 meta	`experiments/results/b3_training_meta.json`
Table II JSON	`experiments/results/table_ii.json`
Calibration CSV	`experiments/results/calibration_val.csv`

8 Conclusion

9 Conclusion

We presented Edge-IQA and LangGraph confidence-aware routing for robotic TCM tongue-image acquisition on a Franka ROS 2 stack, validated end-to-end on the public ShezhenV3-COCO corpus (6,719 images). Keeping IQA and routing outside the 1 kHz control loop preserves real-time safety while enabling JSONL-audited closed-loop policies. On the held-out test split with real Edge-IQA scores, baseline B2 improves median RTT and valid-frame rate versus naive upload; τ ablation and validation-split calibration (τ=0.498, M5 ρ=0.47) characterize operating points under real tongue texture.

Limitations. Validation blur is synthetically injected; real robot motion profiles may differ. Edge-IQA overlap between clear and blurred cohorts on real images indicates room for ROI cropping or learned refinement (B3 explores this direction with a simulated boost only). Cloud vision and EMR integration remain stubs; M6 confirms execution-stack feasibility but does not replace offline RTT statistics.

Future work. Tongue-bbox–conditioned scoring, Gazebo camera injection, train-split fine-tuning for B3 with frozen main-table protocols, and multi-center clinical acquisition studies.

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A Supplementary Materials

Reproduction one-liner. From repository root: `bash scripts/paper1_run_tcm_full.sh` imports ShezhenV3 splits, trains B3 MobileNet, calibrates τ with ROI, runs the main matrix (3 seeds), rebuilds figures/tables, and compiles PDFs.

Artifact inventory.

- `experiments/splits/tcm_{train,val,test}.json` — 6,719 images with COCO `bboxes`;
- `experiments/results/recommended_tau.json` — ROI calibration (n=1,144);
- `experiments/results/roi_ablation.json` — full-frame vs. ROI comparison;
- `experiments/results/b3_mobilenet.pt` — learned B3 checkpoint;
- `experiments/results/b3_training_meta.json` — validation accuracy and sample counts;
- `experiments/results/calibration_val.csv` — per-image Edge-IQA scores;
- `experiments/results/main_seed{0,1,2}_detail.csv` — frame-level matrix logs;
- `figures/fig_iqa_hist.pdf`, `fig_main.pdf` — histogram and bar charts.

Environment. Edge-IQA requires NumPy and Pillow only. B3 training uses PyTorch/torchvision in `doctor/paper1/.venv` (see `train_b3_mobilenet.py`). WSL users mount ShezhenV3 at `/mnt/d/.../shezhenv3-coco` per `tcm_paths.yaml`.

Hyperparameter log. Edge-IQA: 512 resize, 0.65/0.35 sharpness/exposure weights, Laplacian scale 0.002. ROI: 8% padding on union COCO box. B3: MobileNetV3-Small, 224 input, AdamW 10^{-3}, 5 epochs, batch 64, clear vs. blur ($r \in [2.5, 5.0]$). Routing: τ from validation medians, $K=2$, latency model unchanged from phase-2.

JSONL schema (10 fields). `trial_id`, `image_path`, `q_img`, `flags`, `retry_count`, `route_decision`, `latency_ms`, `t_capture`, `t_iqa`, `t_route`.

Ablation scripts. `run_ablation_tau.py` sweeps τ; `roi_ablation.py` compares full vs. ROI Edge-IQA; `cross_dataset_report.py` computes M5 Spearman.

Ethics. ShezhenV3-COCO is used under its public license for algorithm validation; no new human subjects were recruited for Paper I offline experiments.

Limitations (supplementary). (1) Blur labels are synthetic Gaussian, not clinical motion traces. (2) ROI requires detector or dataset boxes at runtime. (3) B3 lacks exposure flags unless hybridized with Edge-IQA. (4) M6 Franka RTT uses a different latency stack than Table II.

Future checklist. Integrate on-device tongue detector for ROI when COCO boxes absent; joint train multi-task blur+exposure head; federated τ calibration across clinics.

B Extended Algorithms and Tables

Per-seed Table II breakdown. Seed 0: B2 M1 p50= 211.1 ms, M2= 0.610; B3 M2= 0.944. Seed 1: B2 M1 p50= 205.6 ms, M2= 0.602; B3 M2= 0.938. Seed 2: B2 M1 p50= 208.3 ms, M2= 0.600; B3 M2= 0.920. Variance across seeds is modest relative to B0–B1 gaps.

Algorithm 3 COCO ROI crop (union + padding)

Input: image I, boxes $\{b_j = [x, y, w, h]\}$, padding p $B \leftarrow \text{union}(\{b_j\})$ Expand B by fraction p on each side; clamp to image bounds $\text{crop}(I, B)$

Algorithm 4 B3 inference (MobileNetV3-Small)

Input: RGB image, optional boxes $I' \leftarrow \text{ROIcrop}(I, \text{boxes})$ $x \leftarrow \text{normalize}(\text{resize}(I', 224))$ $p \leftarrow \text{softmax}(\text{MobileNetV3}(x))$ $Q_{\text{img}} \leftarrow p_{\text{clear}}$

Import statistics. Train 5594, val 572, test 553; 5404/5594 train images carry at least one COCO box.

File checksums (operational). Record SHA256 of `b3_mobilenet.pt` and `recommended_tau.json` in lab notebooks when freezing a submission tag.

Gateway environment variables. `PAPER1_MODE`, `PAPER1_ROOT`, `PAPER1_IQA_SUBPROCESS`, `PAPER1_VISION_THRESHOLD_TAU`—see `docs-zh/paper1/V19/ENV.md`.

Extended related datasets. Other tongue corpora (without robotic acquisition metadata) may fine-tune B3 but require new τ calibration before routing claims transfer.

Glossary. **M1** end-to-end RTT; **M2** valid upload rate; **M3** retry rate; **M5** Spearman correlation between Q_{img} and clear/blur label.

Contact for artifacts. Issue tracker and `REPRODUCE.md` in `doctor/paper1/`.

Table 19: B3 training hyperparameters (ShezhenV3 train).

Hyperparameter	Value
Architecture	MobileNetV3-Small (ImageNet init)
Input size	224×224
Optimizer	AdamW, lr 10^{-3}, wd 10^{-4}
Batch size	64
Epochs	5
Train pairs	11,188 (5594 clear + 5594 blur)
Blur radius	$\mathcal{U}(2.5, 5.0)$
ROI padding	8%
Val accuracy	98.34%

Table 20: Latency model components (milliseconds).

Component	Value
Capture	5.0 (fixed)
Edge-IQA	$\mathcal{N}(9, 2^2)$
Route decision	1.5
Resample (lognormal)	p50 $\approx$50
Cloud upload	$\mathcal{U}(50, 550)$
B3 extra (approx.)	+15–40 on CPU